UPSA
UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA
COMMUNICATION STUDIES

LOGIC AND CRITICAL THINKING

UNIT FIVE: MEANING AND DEFINITIONS

Note: synopsis in this unit were picked from the reading materials below (references).

2025/2026

OUTLINE OF UNIT

a. THE PLACE OF DEFINITIONS IN CRITICAL THINKING
b. WHAT IS DEFINITION?
c. MEANING IN LANGUAGE
d. CATEGORIES OF MEANING
    - Extensional definition
    - Intensional definition
e. THE NEED FOR PRECISION

THE PLACE OF DEFINITIONS IN CRITICAL THINKING

- **Firstly**, knowing how to **construct and evaluate definitions** is fundamental to attaining **the goals of critical thinking** itself.
- If we are trying to decide whether to accept or believe some claim or statement, then we need to make sure that we fully **understand** what the claim or **statement means**.

- **Second**, knowing how to **construct and evaluate definitions** is fundamental to **understanding** and fully engaging in **an academic discipline**. The concepts that are basic to Biology, for instance, are very different from those that are basic to Chemistry or Physics.

- **Finally**, knowing how to **construct and evaluate a definition** is often a fundamental step in **solving problems**.

WHAT THEN IS DEFINITION?

- We may define ***definition*** as a group of words that assigns a **meaning** to some word or group of words.
- Definition refers to the meaning of **symbols** (not of objects), because **only symbols** have the meanings that definitions may explain.
- To explain, we can define the ***word*** **"chair"** ***because it has meaning; but a chair itself we cannot define.***
- We can sit on a chair, or paint it, or burn it, or describe it—but we cannot define it because ***an actual chair is not a symbol that has a meaning to be explained.***
- Sometimes we say, misleadingly, that the thing is being defined; in fact, **what we define are always symbols.**

- Two commonly used technical terms are useful in discussing definitions: **definiendeum** and **definiens.**
- The **definiendum** is ***the symbol being defined***.
- The **definiens** *is the symbol (or group of symbols) used to explain the meaning of the definiendum*.
- Put otherwise, the **definiendum** is the term to be defined and the definiens is the definition of it.
- However, it would be a **mistake** to say that the **definiens** is the meaning of the **definiendum**—rather, ***it is another symbol (or group of symbols) that has the same meaning as the definiendum.***

MEANING IN LANGUAGE

- For our purpose, two linguistic functions of language are particularly important:

  (1) to convey information

  (2) to express or evoke feelings.

- Consider, for example, the following statements….

The death penalty, which is legal in thirty-two states, has been carried out most often in Georgia; however, since 1977 Texas holds the record for the greatest number of executions.

The death penalty is a cruel and inhumane form of punishment in which hapless prisoners are dragged from their cells and summarily slaughtered only to satiate the bloodlust of a vengeful public.

- The first statement is intended primarily to **convey information** about the death penalty, whereas the second is **intended to persuade** us that the death penalty is bad.

- The second **accomplishes this function by engaging our feelings** and not, as in an argument, by establishing the truth of a claim.
- Terminology that **conveys information is said to have cognitive meaning**, and terminology that **expresses or evokes feelings is said to have emotive meaning.**
- When a passage or statement is used to **express emotive feelings and attitudes** (***like using words like 'cruel, inhumane, bloodlust, vengeful'***), then it has emotive meaning in addition to its cognitive meaning.
- Critical thinking requires the **use of statements that have cognitive meaning.**
- Critical thinking obliges us to be able to **distinguish between cognitive meaning and emotive meaning.**
- Loaded language with expression could interfere with our **critical acumen.**

CATEGORIES OF MEANING

- The basic units of any ordinary language are words.
- Our main concern in this section, however, is not with words in general but with **terms**.
- A term is any word or arrangement of words that may serve as the subject of a statement.
- We focus on general **terms—terms that are applicable to more than one object**—which are of **critical importance** in reasoning.
- Examples: man, planet, chair etc.

- The **cognitive meaning** of terms comprises two kinds: **intensional and extensional.**
- The **intensional meaning**, or **intension**, consists of the **qualities or attributes** that the term connotes, and the **extensional meaning**, or extension, consists **of the members of the class that the term denotes**.
- For example, the **intensional meaning of the term "cat"** consists of **the attributes of being furry, of having four legs, of moving in a certain way, of emitting certain sounds, and so on**, while **the extensional meaning** consists **of cats themselves—all the cats in the universe.**
- **The term connotes the attributes and denotes the cats.**

- The **intensional meaning** of a term is otherwise known as the **connotation**, and the **extensional meaning** is known as the **denotation**.
- **Intension and extension** are roughly equivalent to the more modern terms; **sense and reference**, respectively.
- Also, note that Logic uses the terms connotation and denotation differently from the way they are used in grammar.
- In Grammar, **connotation refers to the subtle nuances of a word, whereas denotation refers to the word's direct and specific meaning**.

EXTENSIONAL DEFINITION

It consists **of members of the class that the term denotes.**

A. Demonstrative (ostensive) definitions are probably **the most primitive form of definition.**

- All one needs to do is to **point to the item/thing/person in an attempt to define**.
- As the following examples illustrate, such definitions may be either partial or complete, depending on whether all or only some of the members of the class denoted by the definiendum are pointed to:

"Chair" means this and this and this—as you point to several chairs, one after the other.

B. Enumerative definitions assign a meaning to a **term by naming the members of the class the term denotes**. Like demonstrative definitions, they may also be either partial or complete. Example:

"Lecturer" means a person such as Jacob, Ziem, Horsu, Sadat, Awute, Anim….

C. A definition by subclass assigns a meaning to **a term by naming subclasses of the class denoted by the term**.

- Such a definition, too, may be either partial or complete, depending on whether the subclasses named, when taken together, include all the members of the class or only some of them.
- For examples: **"a car"** means Lamborghini, BMW, Ferrari, Tico,..

INTENSIONAL DEFINITION

- **An intensional definition** is one that assigns a meaning to a word by indicating **the qualities or attributes** that the word connotes.

A. A synonymous definition is one in which **the definiens** is a single word that connotes the same attributes as the **definiendum**.

- **In other words, the definiens is a synonym of the word being defined. For example: "Physician" means a Doctor.**

B. An etymological definition assigns a meaning to a word by disclosing the word's ancestry in both its own language and other languages. **For example**: "Philosophy" is from the Greek words Philos (love) and Sophia (wisdom).

C. An operational definition assigns a meaning to a word **by specifying certain experimental procedures** that determine **whether or not the word applies to a certain thing**.

❖ Examples:

One substance is "harder than" another if and only if one scratches the other when the two are rubbed together.

"Brain activity" means that an electroencephalograph shows oscillations when attached to a patient's head.

D. A definition by genus and difference assigns a meaning to a term by identifying **a genus term** and one or more difference words that, when combined, convey the meaning of the term being defined.

- Definition by genus and difference is more generally applicable and achieves more adequate results than any of the other kinds of intensional definition.
- **To explain how it works, we must first explain the meanings of the terms: genus, species, and specific difference.**
- In Logic, ***genus and species*** have a somewhat different meaning than they have in Biology.
- In Logic, **genus** simply means a relatively **larger class, and species means a relatively smaller subclass of the genus.**
- ***The specific difference***, or ***difference***, is the **attribute or attributes that distinguish the various species within a genus.**

SPECIES	MEANS	DIFFERENCE (DIFFERENTIA)	GENUS
Skyscraper	***means***	***very tall***	***Building***
Ice	***means***	***frozen***	***Water***
Fawn	***means***	***very young***	***Deer***

NOTE

- **Definition by genus and difference is the most effective of the intensional definitions for producing the five kinds of definition.**
- **The five kinds are as follows:**

1. Persuasive Definitions is a kind of definition in which an arguer defines a **term in an effort to persuade a reader or listener to agree with the arguer's point of view** regarding the thing being defined.

- Persuasive definitions usually **contain emotional appeals and slanted terms and are often given** in arguments over highly charged political and social topics on which people have firm views.

2. Stipulative Definitions

- A **definition that has a meaning that is deliberately assigned** to some symbol is called a stipulative definition. It is subjective.
- One who introduces a new symbol is free to assign to it, or stipulate, whatever meaning .
- **Even an old term put into a new context may have its meaning stipulated.**
- Definitions of this sort are sometimes called nominal.
- **Stipulative definitions are neither true nor false** and **you will not find them in a dictionary because they are the subjective views of the speaker.**

3.

Lexical Definitions

- **Less personal definitions include lexical definitions and precising definitions.**
- **In a lexical definition, a word is defined in the way it is standardly used in the language.**
- **In other words, the purpose of a lexical definition is to state the conventional, dictionary meaning of a word.**
- **A lexical definition reports a meaning the definiendum already has.**
- That report may be **correct or incorrect**—and therefore it is clear that a **lexical definition may be either true or false**.

4.

Precising Definitions

- A **precising definition** is **intended to make a vague word more precise so that the word's meaning** is not left to the interpretation of the reader or listener.
- In giving a **precising definition**, we should be careful to **avoid attaching fanciful, biased, or purely personal qualities** to the definition.
- Moreover, **a precising definition must be appropriate for the particular context.**

5.

Theoretical Definitions

- They originate **from specific disciplines** involved in researching the definiendum.
- It is a definition of a term **that appears to formulate a theoretically adequate or scientifically useful description of the objects to which the term applies**.
- Theoretical definitions are **the products of our comprehensive understanding in some sphere**.
- **In discussions of this kind, we may find it useful to define "water" as H 2 O, or "energy" as MC2.**
- Another example is the definition of “light” as a form of electromagnetic radiation

THE NEED FOR PRECISION

- To communicate clearly, to defend our claims without confusing or misleading anyone, and to assess the truth of premises presented to us in the arguments of others, we must **insist that language in the context of argumentation be clear and precise**.

- Thinking critically and arguing effectively often depend on recognizing imprecise language—that is, language that is **vague, overgeneral, or ambiguous**.

Vagueness

- One kind of imprecision in language is vagueness. **A word (or group of words) is vague when its meaning is fuzzy and inexact**, so that it's **unclear** in some contexts whether the expression applies to a certain thing or not.
- Generally speaking, however, **vagueness should be avoided** because it frustrates clear thinking and communication.

Overgenerality

- **Vagueness is often confused with overgenerality**. There is, however, an important distinction between the two. **Words are vague if they have fuzzy or inexact boundaries and hence give rise to debatable borderline cases**. In contrast, **words are overgeneralized if the information they provide is too broad and unspecific in a given context.**

- Sometimes, of course, words may be both vague and overgeneralized.
- Thus, if I describe my lost luggage simply as "a large black bag," my description is too vague ("large," "black," and "bag" all have fuzzy, inexact meanings) and too general (the phrase "large black bag" isn't specific enough to distinguish my bag from many others).

Ambiguity

- Ambiguity refers to **a doubtful sense of a word or phrase**. Many words have more than one meaning.
- The word star, for example, can mean, among other things, a Hollywood celebrity or a twinkling celestial object.
- A word or expression **is ambiguous if it has two or more distinct meanings and the context does not make clear** which meaning is intended.

- Ambiguity is what makes puns and many jokes funny, but when used unintentionally it can destroy the effectiveness of an argument.

- **Ambiguity, like overgeneralization, is often confused with vagueness.**

- **The basic difference between ambiguity and vagueness is this:**

- **A vague word is imprecise because it has blurry boundaries and unclear borderline applications.**

- **An ambiguous word is imprecise because it is unclear which of two or more distinct meanings (each of which may be quite precise) is the one intended by the author.**

LET'S ENGAGE

- **Turn to Unit Five of your workbook and answer the questions on:**
- **The Place of Definitions in Critical Thinking and Meaning in Language**
- **Now let us discuss the answers. Exchange your workbook with your partner for marking**

ASSIGNMENT

- **Complete the tasks on Extensional and Intensional Definition and submit it next week**

References

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